

HPI Bridge Design Specification

Ver 0.2

DM270

Department: WW IMAGING

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HISTORY

Version	Date	Author	Notes
Draft	03/19/02	Masaki Kato	Draft for DM270.
Ver0.1	04/08/02	Masaki Kato	Added external memory select bit.
Ver0.2	06/03/02	Masaki Kato	Fixed ARM address. Added DON description.

CONTENTS

1. ABSTRACT	5
2. REFERENCES	5
3. GLOSSARY	5
4. FUNCTION.....	6
4.1 BLOCK DIAGRAM	7
4.2 TIMING.....	7
4.3 STATE DIAGRAM.....	7
5. REGISTERS.....	8
5.1 DETAILED REGISTER	8
6. INTERFACE	12

TABLES

Table 1 : HPIB ARM Register List 8

Table 2 : HPIB DSP Register List..... 8

Table 3 : HPIB Top-level I/O List 12

FIGURES

Figure 1 : HPIB Block Diagram..... 7

1. ABSTRACT

This documentation is described about the HPI Bridge module (HPIB) for DM270. HPIB connects BUSC(ARM bus controller) to Host Port Interface of DSP(SC_DM270DS), and enables ARM to read/write data from/to DSP.

2. REFERENCES

TMS320C54x DSP reference set

3. GLOSSARY

HPIB	Host Port Interface Bridge
DSP	SC_DM270DS subchip
ARM	ARM71TDMIE
HPI8	Host Port Interface using 8bit bus between HPIB module and DSP
HPI16	Host Port Interface using 16bit bus between HPIB module and DSP
BUSC	ARM Bus Controller module

4. FUNCTION

[summary]

HPIB enables ARM to access DSP on-chip RAM.

A side of HPIB module is connected to HPI ports of DSP and the other sides connected to BUSC(ARM bus controller) module. The data between BUSC and this module has 16bit width (minimum 1 wait). But, DSP have 2 type of HPI. One is HPI16 and the other is HPI8. The number following 'HPI' means bus width. DSP have 16bit bus for only HPI16, then simultaneously HPI16-bus and XIO-bus can be used.

HPI8 function is not implemented in this HPIB module.

BUSC accesses this module with one-wait. DSP have HRDY pin, and this module takes this signal, and if the level of it is '0' (DSP is busy), stop to write data to DSP until it rise to '1'. In such case, the write cycle takes more time.

This module is synchronized with ARM clock, and have state machine. And, the slower DSP clock frequency is, the longer it takes time for HRDY to turns to '1'. Then one way to get faster access is to raise both DSP and ARM clock frequencies.

[HPI16 mode]

HPI16 is the default mode, and this module takes address and data in a same time from BUSC in write mode. This module makes strobe signal (hpib_hds1n), HPI chip select signal (hpib_hcsn). When writing data to DSP, BUSC specifies the address, and the data. If the previous operation is not finished, this module set the wait signal to '0'. In reading, this module set the wait signal '0' too, until the data from DSP is valid.

[other signal to DSP]

ARM can control 'DSP reset' and 'DSP hold mode' by writing a specific bit of "HPIB_CTRL register" in this module. BUSC can access "HPIB_CTRL register" without wait..

DSC27での機能変更点

- DMA追加。DMA制御はDSPのIO空間レジスタで行う。
- DMAはシングル転送を使用。4バイト単位での転送のみ可能とする。
- 外部ホストインターフェース追加。
- ARMおよびDMA, 外部ホストの3系統の調停回路搭載。同時アクセス可能。優先度は上位から外部ホスト, DMA, ARMの順。
- DSPメモリ空間拡張(64kワード=128kバイト)に伴い, ARMアドレス1ビット追加。

4.1 Block Diagram

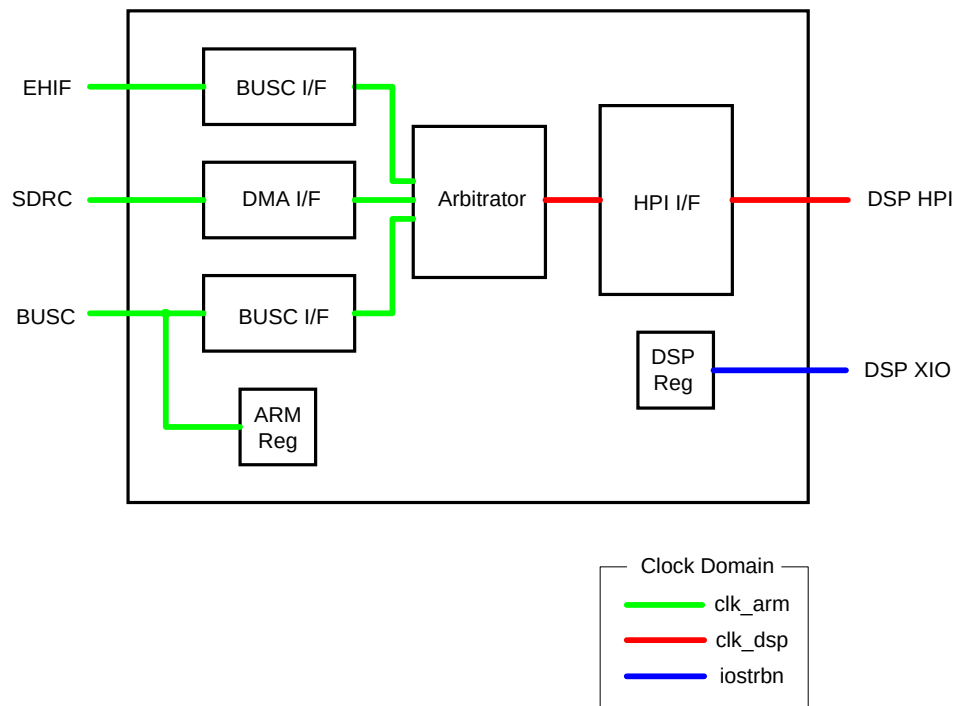


Figure 1 : HPIB Block Diagram

4.2 Timing

4.3 State Diagram

5. REGISTERS

Table 1 : HPIB ARM Register List

Register Name	Offset address	Description
HPIB_CTRL	0x00	HPIB mode select / DSP reset / DSP hold mode select / Interrupt to DSP
HPIB_STAT	0x01	DSP status

Table 2 : HPIB DSP Register List

Register Name	DSP IO address	Description
DMASAD0	0x0080	DMA開始外部メモリ下位アドレス
DMASAD1	0x0081	DMA開始外部メモリ上位アドレス
DMADAD0	0x0082	DMA開始DSP下位アドレス
DMADAD1	0x0083	DMA開始DSP上位アドレス
DMASZ	0x0084	DMA転送サイズ
DMACTL	0x0085	DMA制御
DMATRG	0x0086	DMA転送開始
DMAREST	0x0087	DMA転送残量

5.1 Detailed Register

ARM Register

HPIB_CTRL (0003:0600) offset:0x00

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	DBIO	DHOLD	DRST	DINT0	RSV	EXCHG	RSV	HPNMI	RSV	RSV	HPIEN
R/W	-	-	-	-	-	R/W	R/W	R/W	R/W	-	R/W	-	R/W	-	-	R/W
Default	-	-	-	-	-	1	1	1	1	-	0	-	1	-	-	1

Bits	Register Name	Description
15:11	RSV	Reserved bits
10	DBIO	DSP Branch Control . Connected to BIO terminal of DSP. When DSP is in BIO status, it can branch and control program. 0 : DSP executes the conditional instruction 1 : No operation
9	DHOLD	DSP hold request. Holds DSP. Connected to HOLDN of DSP. 0 : Hold 1 : Normal
8	DRST	Resets DSP. Connected to RSN terminal of DSP. 0 : DSP reset 1 : Normal
7	DINT0	Signal input into INT0 port of DSP. Initial value is 1. INT0 is generated in DSP by the host setting this bit to 0. After interrupt is generated, the host must return this bit to 1. 0 : interrupt (INT0) 1 : Normal
6	RSV	Reserved bit

Bits	Register Name	Description
5	EXCHG	Exchange Upper byte for Lower. This switches the upper byte and lower byte of the data in HPI16 mode. The initial value is 0. When set to 1, byte switching is performed. 0 : / HOST -> DSP: The MSB of 16-bit data passed from the host is written as is in correspondence with the MSB of the internal DARAM of the DSP. 0 : / DSP -> HOST: 16-bit data of the internal DARAM of DSP is passed as is to the host. 1 : / HOST -> DSP: The lower byte of 16-bit data passed from the host is written into the upper byte of the internal DARAM of DSP, and the upper byte is written into the lower byte of DARAM. 1 : / DSP -> HOST: The lower byte of the internal DARAM of DSP is written into the upper byte of the bus to the host, and the upper byte is written into the lower byte of the bus.
4	RSV	Reserved bit
3	HPNMI	HPI Non Mask able Interrupt. Host generates NMI in DSP. Connected to NMI port of DSP. After interrupt is generated, it must be reset to normal status. 0 : interrupt (NMI) 1 : Normal
2:1	RSV	Reserved bit
0	HPIEN	Enables HPI. Connected to HPIEN of DSP. When disabled, HPI function are invalid. 0 : Disable 1 : Enable

HPIB_STAT (0003:0602) Offset:0x01

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	XF	RSV	RSV	RSV	HOLDA	RSV	RSV	RSV	RSV	RSV	RSV	RSV	HINT
R/W	-	-	-	R	-	-	-	R	-	-	-	-	-	-	-	R
Default	-	-	-	1	-	-	-	1	-	-	-	-	-	-	-	1

Bits	Register Name	Description
15:13	RSV	Reserved bits
12	DXF	DSP XF indicator. Shows status o DSP XF signal output.
11:9	RSV	Reserved bits
8	HOLDA	DSP Hold Acknowledge. Indicates status of DSP HOLDA signal output. After a hold request is sent to DSP by the DHOLD bit of the HIB_CTRL register, acknowledgement can be checked by this bit. 0 : DSP hold status 1 : Normal
7:1	RSV	Reserved bits
0	HINT	Host port interrupt. Indicates status of DSP HINT signal output. 0 : DSP sends interrupt request to host 1 : Normal

DSP Register**DMASAD0 (0x0080)**

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	SAD15	SAD14	SAD13	SAD12	SAD11	SAD10	SAD9	SAD8	SAD7	SAD6	SAD5	SAD4	SAD3	SAD2	SAD1	SAD0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R	R
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	SAD15-SAD0	外部メモリのDMA転送開始下位アドレス。4バイト単位でのアドレスを設定してください。下位2ビットは0固定です。

DMASAD1 (0x0081)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	SAD21	SAD20	SAD19	SAD18	SAD17	SAD16
R/W	-	-	-	-	-	-	-	-	-	-	R/W	R/W	R/W	R/W	R/W	R/W
Default	-	-	-	-	-	-	-	-	-	-	0	0	0	0	0	0

Bits	Register Name	Description
15:6	RSV	Reserved
5:0	SAD21-SAD16	外部メモリのDMA転送開始上位アドレス。

DMADAD0 (0x0082)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DAD15	DAD14	DAD13	DAD12	DAD11	DAD10	DAD9	DAD8	DAD7	DAD6	DAD5	DAD4	DAD3	DAD2	DAD1	DAD0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	DAD15-DAD0	DSPのDMA転送開始下位アドレス。DSPワードアドレスを指定して下さい。

DMADAD1 (0x0083)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	DAD16
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Register Name	Description
15:1	RSV	Reserved
0	DAD16	DSPのDMA転送開始上位アドレス。DSPワードアドレスを指定して下さい。

DMASZ (0x0084)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DSZ15	DSZ14	DSZ13	DSZ12	DSZ11	DSZ10	DSZ9	DSZ8	DSZ7	DSZ6	DSZ5	DSZ4	DSZ3	DSZ2	DSZ1	DSZ0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R	R
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	DSZ15-DSZ0	DMA転送バイト数。4バイト単位で指定して下さい。下位2ビットは0固定です。

DMACTL (0x0085)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	ENDI1	ENDI0	RSV	RSV	RSV	MEM	RSV	RSV	RSV	DIR
R/W	-	-	-	-	-	-	R/W	R/W	-	-	-	R/W	-	-	-	R/W
Default	-	-	-	-	-	-	0	0	-	-	-	0	-	-	-	0

Bits	Register Name	Description
15:10	RSV	リザーブ。
9:8	ENDI1-ENDI0	エンディアン設定。 0 : リトルエンディアン 1 : ビッグエンディアン 2 : バイトスワップ
7:5	RSV	リザーブ。
4	MEM	外部メモリ選択。 0 : SDRAM 1 : フラッシュメモリ
3:1	RSV	リザーブ。
0	DIR	DMA転送方向。 0 : 外部メモリ → DSP 1 : 外部メモリ ← DSP

DMATRG (0x0085)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	RSV	TRG
R/W	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	R/W
Default	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0

Bits	Register Name	Description
15:1	RSV	リザーブ。
0	TRG	DMA転送トリガ。1を書き込むとDMA転送がスタートします。転送中に0を書くと転送が中止されます。転送中は1が読め、転送が終了すると自動的に0になります。1の書きこみから転送ステータスのセットまでは数サイクルかかりますのでポーリング等を行う場合は注意が必要です。

DMAREST (0x0086)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	DREST15	DREST14	DREST13	DREST12	DREST11	DREST10	DREST9	DREST8	DREST7	DREST6	DREST5	DREST4	DREST3	DREST2	DREST1	DREST0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
Default	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bits	Register Name	Description
15:0	DREST15-DREST0	DMA転送残量。転送中に現在のDMA転送残量が示されます。

6. Interface

Table 3 : HPIB Top-level I/O List

Pin Name	Bits	I/O	Fr/To	Description
Reset / clock in normal mode				
rst_n	1	I	EXT	Power on reset
clk_arm	1	I	CLKC	ARM clock
clk_dsp	1	I	CLKC	DSP clock
BUSC interface				
busc_a	16	I	BUSC	Address(half-word) bus from ARM
busc_do	16	I	BUSC	Data bus from ARM
busc_ncs_hpib	1	I	BUSC	Chip select (DSP memory access)
busc_ncs_dspc	1	I	BUSC	Chip select (HPIB register access)
busc_nrw	1	I	BUSC	Read_/Write signal
hpib_do	16	O	BUSC	Data bus to ARM (DSP memory access)
dspc_do	16	O	BUSC	Data bus to ARM (HPIB register access)
hpib_nwait	1	O	BUSC	Wait request signal to ARM
brdg_lock	1	I	BRDG	Locked access signal
External Host IF interface				
ehif_a	16	I	HRCIF	Address bus from EHIF
ehif_do	16	I	HRCIF	Data bus from EHIF
ehif_ncs_hpib	1	I	HRCIF	Chip select (DSP memory access)
ehif_nrw	1	I	HRCIF	Read_/Write signal
hpib_do_eh	16	O	HRCIF	Data bus to EHIF (DSP memory access)
hpib_nwait_eh	1	O	HRCIF	Wait request signal to EHIF
DSP interface (HPI)				
dsp_h16do	16	I	DSP	HPI16 input data bus
dsp_hdo	8	I	DSP	HPI8 input data bus (no use internally)
dsp_hrdy	1	I	DSP	HPI16 access ready signal
hpib_ai	17	O	DSP	HPI16 address bus
hpib_hdi	16	O	DSP	HPI16 output data bus
hpib_hcsn	1	O	DSP	HPI16 chip select
hpib_hds1n	1	O	DSP	HPI16 data strobe
hpib_hrnw	1	O	DSP	HPI16 read/write_ signal
hpib_hpiena	1	O	DSP	HPI16 HPI function enable
hpib_hpi16	1	O	DSP	HPI16 mode select signal (tied off high internally)
hpib_hcntl0	1	O	DSP	HPI8 host access control 0 (tied off high internally)
hpib_hcntl1	1	O	DSP	HPI8 host access control 1 (tied off high internally)
hpib_hbil	1	O	DSP	HPI8 byte identification (tied off low internally)
DSP interface (INTERRUPT)				
dsp_hintn	1	I	DSP	Interrupt signal from DSP
dspc_int0n	1	O	DSP	interrupt to DSP (INT0N)
dspc_nmin	1	O	DSP	interrupt to DSP (NMI)
hpib_dsp_int	1	O	DSP	interrupt to DSP (INT4N)
DSP interface (XIO)				
dsp_istrob	1	I	CLKC	DSP I/O strobe
dsp_is	1	I	DSP	DSP I/O space select
dsp_rw	1	I	DSP	DSP read/write_
dsp_a	16	I	DSP	DSP address bus
dsp_do	16	I	DSP	DSP output data bus
hpib_dsp_di	16	O	DSP	DSP input data bus
DSP interface (others)				

Pin Name	Bits	I/O	Fr/To	Description
dsp_holdan	1	I	DSP	DSP hold mode acknowledge signal
dsp_xf	1	I	DSP	DSP XF output
hpib_rsn	1	O	DSP	DSP reset signal
hpib_bion	1	O	DSP	DSP branch control
hpib_holdn	1	O	DSP	Hold mode request to DSP
DMA interface				
hpib_dma_req	1	O	SDRC	Memory request
hpib_dma_addr	27	O	SDRC	SDRAM address
hpib_dma_rnw	1	O	SDRC	Read/write
hpib_dma_bcmt	6	O	SDRC	DMA byte count
hpib_dma_do	32	O	SDRC	Write data to SDRAM
sdrc_rdy_hpib	1	I	SDRC	Request enable
sdrc_rdata_dma	32	I	SDRC	Read data from SDRAM
OTHER FUNCTIONAL PORT(S)				
hpib_nint	1	O	INTC	ARM interrupt pulse generated by DSP HINT

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